



HTML Notes-anisha

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HTML:

HTML stands for Hyper Text Markup Language, which is the most widely used language on Web to develop web pages. HTML was created by Berners-Lee in late 1991 but "HTML 2.0" was the first standard HTML specification which was published in 1995. HTML 4.01 was a major version of HTML and it was published in late 1999. Though HTML 4.01 version is widely used but currently we are having HTML-5 version which is an extension to HTML 4.01, and this version was published in 2012.

HTML stands for **H**ypertext **M**arkup **L**anguage, and it is the most widely used language to write Web Pages.

- **Hypertext** refers to the way in which Web pages (HTML documents) are linked together. Thus, the link available on a webpage is called Hypertext.
- As its name suggests, HTML is a **Markup Language** which means you use HTML to simply "mark-up" a text document with tags that tell a Web browser how to structure it to display.

Originally, HTML was developed with the intent of defining the structure of documents like headings, paragraphs, lists, and so forth to facilitate the sharing of scientific information between researchers.

Now, HTML is being widely used to format web pages with the help of different tags available in HTML language.

BASIC HTML DOCUMENT

```
<html>

<head>

  <title>This is document title</title>

</head>

<body>

  <h1>This is a heading</h1>

  <p>Document content goes here.....</p>

</body>
```

</html>

HTML TAGS

As told earlier, HTML is a markup language and makes use of various tags to format the content. These tags are enclosed within angle braces **<Tag Name>**. Except few tags, most of the tags have their corresponding closing tags. For example, **<html>** has its closing tag **</html>** and **<body>** tag has its closing tag **</body>** tag etc.

Sr.No	Tag & Description
1	<!DOCTYPE...> This tag defines the document type and HTML version.
2	<html> This tag encloses the complete HTML document and mainly comprises of document header which is represented by <head>...</head> and document body which is represented by <body>...</body> tags.
3	<head> This tag represents the document's header which can keep other HTML tags like <title> , <link> etc.
4	<title> The <title> tag is used inside the <head> tag to mention the document title.
5	<body> This tag represents the document's body which keeps other HTML tags like <h1> , <div> , <p> etc.
6	<h1> This tag represents the heading.
7	<p>

	This tag represents a paragraph.
--	----------------------------------

HTML DOCUMENT STRUCTURE

A typical HTML document will have the following structure –

```
<html>
```

```
<titile>
```

```
KICE
```

```
</titile>
```

```
<body>
```

```
<h1>This is a heading</h1>
```

```
<p>Document content goes here.....</p>
```

```
</body>
```

```
</html>
```

HEADING TAGS

Any document starts with a heading. You can use different sizes for your headings. HTML also has six levels of headings, which use the elements **<h1>**, **<h2>**, **<h3>**, **<h4>**, **<h5>**, and **<h6>**. While displaying any heading, browser adds one line before and one line after that heading.

Example:

```
<html>

<head>

  <body>

    <h1>This is heading 1</h1>

    <h2>This is heading 2</h2>

    <h3>This is heading 3</h3>

    <h4>This is heading 4</h4>

    <h5>This is heading 5</h5>

    <h6>This is heading 6</h6>

  </body>

</head>

</html>
```

PARAGRAPH TAG

The **<p>** tag offers a way to structure your text into different paragraphs. Each paragraph of text should go in between an opening **<p>** and a closing **</p>** tag as shown below in the example –

Example

```
<html>

<head>

  <body>

    <p>Whenever you use the  element, anything following it starts from the next line. This tag is an example of an empty element, where you do not need opening and closing tags, as there is nothing to go in between them...</p>

    <p>The  tag has a space between the characters br and the forward slash. If you omit this space, older browsers will have trouble rendering the line break, while if you miss the forward slash character and just use  it is not valid in XHTML..</p>

    <p>you can use  tag to put any content in the center of the page or any table cell..</p>
```

```
</head>

</body>

</html>
```

LINE BREAK TAG

Whenever you use the **
** element, anything following it starts from the next line. This tag is an example of an **empty** element, where you do not need opening and closing tags, as there is nothing to go in between them.

The **
** tag has a space between the characters **br** and the forward slash. If you omit this space, older browsers will have trouble rendering the line break, while if you miss the forward slash character and just use **
** it is not valid in XHTML.

Example

```
<html>

<head>

<body>

  <p>Hello<br />

    You delivered your assignment ontime.<br />

    Thanks<br />

    Mahnaz</p>

</head>

</body>

</html>
```

HORIZONTAL LINES

Horizontal lines are used to visually break-up sections of a document. The **<hr>** tag creates a line from the current position in the document to the right margin and breaks the line accordingly.

For example, you may want to give a line between two paragraphs as in the given example below –

Example

```
<html>

<head>

  <title>Horizontal Line Example</title>

</head>

<body>

  <p>This is paragraph one and should be on top</p>

  <hr />

  <p>This is paragraph two and should be at bottom</p>

</body>
```

Again **<hr />** tag is an example of the **empty** element, where you do not need opening and closing tags, as there is nothing to go in between them.

The **<hr />** element has a space between the characters **hr** and the forward slash. If you omit this space, older browsers will have trouble rendering the horizontal line, while if you miss the forward slash character and just use **<hr>** it is not valid in XHTML

PRESERVE FORMATTING

Sometimes, you want your text to follow the exact format of how it is written in the HTML document. In these cases, you can use the preformatted tag **<pre>**.

Any text between the opening **<pre>** tag and the closing **</pre>** tag will preserve the formatting of the source document.

Example

```
<html>

<head>

  <title>Preserve Formatting Example</title>
```

```
</head>

<body>

  <pre>

    Suppose you want to use the phrase "12 Angry Men." Here, you would not want a browser to split the
    "12, Angry" and "Men" across two lines

  </pre>

</body>

</html>
```

NONBREAKING SPACES

Suppose you want to use the phrase "12 Angry Men." Here, you would not want a browser to split the "12, Angry" and "Men" across two lines –

An example of this technique appears in the movie "12 Angry Men."

In cases, where you do not want the client browser to break text, you should use a nonbreaking space entity ** ** instead of a normal space. For example, when coding the "12 Angry Men" in a paragraph, you should use something similar to the following code –

Example

```
<html>

<head>

  <title>Nonbreaking Spaces Example</title>

</head>

<body>

  <p>An example of this technique appears in the movie "12&nbsp;Angry&nbsp;Men."</p>

</body>

</html>
```

This will produce the following result –

Start Tag	Content	End Tag
<p>	This is paragraph content.	</p>
<h1>	This is heading content.	</h1>
<div>	This is division content.	</div>

So here **<p>...</p>** is an HTML element, **<h1>...</h1>** is another HTML element. There are some HTML elements which don't need to be closed, such as **<img.../>**, **<hr />** and **
** elements. These are known as **void elements**.

HTML TAG VS. ELEMENT

An HTML element is defined by a starting tag. If the element contains other content, it ends with a closing tag.

For example, **<p>** is starting tag of a paragraph and **</p>** is closing tag of the same paragraph but **<p>This is paragraph</p>** is a paragraph element.

NESTED HTML ELEMENTS

It is very much allowed to keep one HTML element inside another HTML element –

Example

```
<html>
  <head>
    <title>Nested Elements Example</title>
  </head>
  <body>
```

```
<h1>This is <i>italic</i> heading</h1>

<p>This is <u>underlined</u> paragraph</p>

</body>

</html>
```

HTML - ATTRIBUTES

We have seen few HTML tags and their usage like heading tags **<h1>**, **<h2>**, paragraph tag **<p>** and other tags. We used them so far in their simplest form, but most of the HTML tags can also have attributes, which are extra bits of information.

An attribute is used to define the characteristics of an HTML element and is placed inside the element's opening tag. All attributes are made up of two parts – a **name** and a **value**

- The **name** is the property you want to set. For example, the paragraph **<p>** element in the example carries an attribute whose name is **align**, which you can use to indicate the alignment of paragraph on the page.
- The **value** is what you want the value of the property to be set and always put within quotations. The below example shows three possible values of align attribute: **left**, **center** and **right**.

Attribute names and attribute values are case-insensitive. However, the World Wide Web Consortium (W3C) recommends lowercase attributes/attribute values in their HTML 4 recommendation.

Example

```
<html>

<head>

  <title>Align Attribute Example</title>

</head>

<body>

  <p align = "left">This is left aligned</p>

  <p align = "center">This is center aligned</p>
```

```
<p align = "right">This is right aligned</p>
</body>
</html>
```

HTML - FORMATTING

BOLD TEXT

Anything that appears within **...** element, is displayed in bold as shown below –

Example

```
<html>
<head>
  <title>Bold Text Example</title>
</head>
<body>
  <p>The following word uses a <b>bold</b> typeface.</p>
</body>
</html>
```

ITALIC TEXT

Anything that appears within **<i>...</i>** element is displayed in italicized as shown below

Example

```
<html>
<head>
  <title>Italic Text Example</title>
</head>
<body>
  <p>The following word uses an <i>italicized</i> typeface.</p>
</body>
```

```
</html>
```

UNDERLINED TEXT

Anything that appears within `<u>...</u>` element, is displayed with underline as shown below –

Example

```
<html>

<head>

  <title>Underlined Text Example</title>

</head>

<body>

  <p>The following word uses an <u>underlined</u> typeface.</p>

</body>

</html>
```

SUPERSCRIFT TEXT

The content of a `<sup>...</sup>` element is written in superscript; the font size used is the same size as the characters surrounding it but is displayed half a character's height above the other characters.

Example

```
<html>

<head>

  <title>Superscript Text Example</title>

</head>

<body>

  <p>The following word uses a <sup>superscript</sup> typeface.</p>
```

```
</body>
</html>
```

SUBSCRIPT TEXT

The content of a **_{...}** element is written in subscript; the font size used is the same as the characters surrounding it, but is displayed half a character's height beneath the other characters.

Example

```
<html>
  <head>
    <title>Subscript Text Example</title>
  </head>
  <body>
    <p>The following word uses a <sub>subscript</sub> typeface.</p>
  </body>
</html>
```

HTML Element

The HTML element defines deleted/removed text.

Example

```
<p>My favorite color is <del>blue</del> red.</p>
```

INSERTED TEXT

Anything that appears within **<ins>...</ins>** element is displayed as inserted text.

Example

```
<html>
  <head>
```

```
<title>Inserted Text Example</title>

</head>

<body>

  <p>I want to drink <del>cola</del> <ins>wine</ins></p>

</body>

</html>
```

SMALLER TEXT

The content of the **<small>...</small>** element is displayed one font size smaller than the rest of the text surrounding it as shown below –

Example

```
<html>

<head>

  <title>Smaller Text Example</title>

</head>

<body>

  <p>The following word uses a <small>small</small> typeface.</p>

</body>

</html>
```

EMPHASIZED TEXT

Anything that appears within **...** element is displayed as emphasized text.

Example

```
<html>

<head>

  <title>Emphasized Text Example</title>

</head>

<body>
```

```
<p>The following word uses an <em>emphasized</em> typeface.</p>
</body>
</html>
```

MARKED TEXT

Anything that appears with-in **<mark>...</mark>** element, is displayed as marked with yellow ink.

Example

```
<html>
<head>
  <title>Marked Text Example</title>
</head>
<body>
  <p>The following word has been <mark>marked</mark> with yellow</p>
</body>
</html>
```

STRONG TEXT

Anything that appears within **...** element is displayed as important text.

Example

```
<html>
<head>
  <title>Strong Text Example</title>
</head>
<body>
  <p>The following word uses a <strong>strong</strong> typeface.</p>
</body>
```

```
</html>
```

GROUPING CONTENT

- <div> element –block level element
- element-inline element

The <div> Element

The <div> element is often used as a container for other HTML elements.

The <div> element has no required attributes, but style, class and id are common.

When used together with CSS, the <div> element can be used to style blocks of content:

Example

```
<div style="background-color:red;color:white;padding:50px;">

  <h2>London</h2>

  <p>London is the capital city of England. It is the most populous city in the United Kingdom, with a metropolitan area of over 13 million inhabitants.</p>

</div>
```

The Element

The element is often used as a container for some text.

The element has no required attributes, but style, class and id are common.

When used together with CSS, the element can be used to style parts of the text:

Example

```
<h1>My <span style="color:red">Important</span> Heading</h1>
```

HTML QUOTATIONS ELEMENT

<abbr> element

The HTML <abbr> element defines an abbreviation or an acronym.

Marking abbreviations can give useful information to browsers, translation systems and search-engines.

Example

```
<p>The <abbr title="World Health Organization">WHO</abbr> was founded in 1948.</p>
```

TEXT DIRECTION

The **<bdo>...</bdo>** element stands for Bi-Directional Override and it is used to override the current text direction.

Example

```
<html>

<head>

  <title>Text Direction Example</title>

</head>

<body>

  <p>This text will go left to right.</p>

  <p><bdo dir = "rtl">This text will go right to left.</bdo></p>

</body>

</html>
```

SPECIAL TERMS <DFN> TAG

The **<dfn>...</dfn>** element (or HTML Definition Element) allows you to specify that you are introducing a special term. It's usage is similar to italic words in the midst of a paragraph.

Typically, you would use the <dfn> element the first time you introduce a key term. Most recent browsers render the content of a <dfn> element in an italic font.

Example

```
<html>
```

```
<head>

  <title>Special Terms Example</title>

</head>

<body>

  <p>The following word is a <dfn>special</dfn> term.</p>

</body>

</html>
```

<q> element

Used to insert quotes i.e. short phrase.

Browsers usually insert quotation marks around the <q> element.

Example

```
<p> Great quote on love and life.</p>

<p> Dr. Seuss once said : <q>Reality is finally better than your dreams.</q></p>
```

<blockquote> element

Used to insert some section that is quoted from another source.

Browsers usually indent <blockquote> elements

Example

```
<p> Read this inspirational story </p>

<blockquote> According to scientists, the bumblebee's body is too heavy and its wing span
too small. Aerodynamically, the bumblebee cannot fly. But the bumblebee doesn't know th
at and it keeps flying. When you don't know your limitations, you go out and surprise your
self. In hindsight, you wonder if you had any limitations. The only limitations a person has a
re those that are selfimposed. Don't let education put limitations on you</blockquote>
```

TEXT CITATIONS

<cite> element

The HTML <cite> element defines the title of a work.

Browsers usually display <cite> elements in italic.

Example

```
<p><cite>The Scream</cite> by Edvard Munch. Painted in 1893.</p>
```

COMPUTER CODE

Any programming code to appear on a Web page should be placed inside **<code>...</code>** tags. Usually the content of the <code> element is presented in a monospaced font, just like the code in most programming books.

Example

```
<html>

<head>

  <title>Computer Code Example</title>

</head>

<body>

  <p>Regular text. <code>This is code.</code> Regular text.</p>

</body>

</html>
```

KEYBOARD TEXT

When you are talking about computers, if you want to tell a reader to enter some text, you can use the **<kbd>...</kbd>** element to indicate what should be typed in, as in this example.

Example

```
<html>

<head>

  <title>Keyboard Text Example</title>
```

```
</head>

<body>

  <p>Regular text. <kbd>This is inside kbd element</kbd> Regular text.</p>

</body>

</html>
```

<address>

The HTML <address> element defines contact information (author/owner) of a document or an article.

The <address> element is usually displayed in italic. Most browsers will add a line break before and after the element.

Example

```
<address>

Written by Joseph.<br>

Visit us at:<br>

joseph.com<br>

57, Anuparpalayam, <br>

</address>
```

HTML IMAGES

- In HTML, images are defined with the tag.
- The tag is empty, it contains attributes only, and does not have a closing tag.

The **src attribute** specifies the URL (web address) of the image:

The **alt attribute** provides an alternate text for an image, if the user for some reason cannot view it (because of slow connection, an error in the src attribute, or if the user uses a screen reader).

Example

```
<!DOCTYPE html>
```

```
<html>

<body>

<h2>Alternative text</h2>

<p>The alt attribute should reflect the image content, so users who cannot see the
image gets an understanding of what the image contains:</p>



</body>

</html>
```

Image Size - Width and Height

You can use the style attribute to specify the width and height of an image.

Example

```
<!DOCTYPE html>

<html>

<body>

<h2>Image Size</h2>

<p>Use the style attribute to specify the width and height of an image:</p>



</body>

</html>
```

Images in Another Folder

Example

```
<!DOCTYPE html>

<html>

<body>

<h2>Images in Another Folder</h2>
```

<p>It is common to store images in a sub-folder. You must then include the folder name in the src attribute:</p>

```

```

```
</body>
```

```
</html>
```

Images on Another Server

Example

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>Images on Another Server</h2>
```

```

```

```
</body>
```

```
</html>
```

Animated Images

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>Animated Images</h2>
```

```
<p>The GIF standard allows moving images.</p>
```

```

```

```
</body>
```

```
</html>
```

Image as a Link

Example

```
<!DOCTYPE html>

<html>

<body>

<h2>Image as a Link</h2>

<p>The image is a link. You can click on it.</p>

<a href="default.asp">

</a>

</body>

</html>
```

HTML - TABLES

The HTML tables allow web authors to arrange data like text, images, links, other tables, etc. into rows and columns of cells.

The HTML tables are created using the **<table>** tag in which the **<tr>** tag is used to create table rows and **<td>** tag is used to create data cells. The elements under **<td>** are regular and left aligned by default

Example

```
<html>

<head>

  <title>HTML Tables</title>

</head>

<body>
```

```

<table border = "1">

  <tr>

    <td>Row 1, Column 1</td>

    <td>Row 1, Column 2</td>

  </tr>

  <tr>

    <td>Row 2, Column 1</td>

    <td>Row 2, Column 2</td>

  </tr>

</table>

</body>

</html>

```

Row 1, Column 1	Row 1, Column 2
Row 2, Column 1	Row 2, Column 2

TABLE HEADING

Table heading can be defined using **<th>** tag. This tag will be put to replace **<td>** tag, which is used to represent actual data cell. Normally you will put your top row as table heading as shown below, otherwise you can use **<th>** element in any row. Headings, which are defined in **<th>** tag are centered and bold by default.

Example

```

<html>

  <head>

    <title>HTML Table Header</title>

  </head>

  <body>

    <table border = "1">

```

```
<tr>

  <th>Name</th>

  <th>Salary</th>

</tr>

<tr>

  <td>Ramesh Raman</td>

  <td>5000</td>

</tr>

<tr>

  <td>Shabbir Hussein</td>

  <td>7000</td>

</tr>

</table>

</body>

</html>
```

COLSPAN AND ROWSPAN ATTRIBUTES

You will use **colspan** attribute if you want to merge two or more columns into a single column. Similar way you will use **rowspan** if you want to merge two or more rows.

Example

```
<html>

<head>

  <title>HTML Table Colspan/Rowspan</title>

</head>

<body>

  <table border = "1">

    <tr>
```

```

<th>Column 1</th>

<th>Column 2</th>

<th>Column 3</th>

</tr>

<tr>

<td rowspan = "2">Row 1 Cell 1</td>

<td>Row 1 Cell 2</td>

<td>Row 1 Cell 3</td>

</tr>

<tr>

<td>Row 2 Cell 2</td>

<td>Row 2 Cell 3</td>

</tr>

<tr>

<td colspan = "3">Row 3 Cell 1</td>

</tr>

</table>

</body>

</html>

```

TABLE HEIGHT AND WIDTH

You can set a table width and height using **width** and **height** attributes. You can specify table width or height in terms of pixels or in terms of percentage of available screen area.

Example

```

<html>

<head>

<title>HTML Table Width/Height</title>

```

```
</head>

<body>

  <table border = "1" width = "400" height = "150">

    <tr>

      <td>Row 1, Column 1</td>

      <td>Row 1, Column 2</td>

    </tr>

    <tr>

      <td>Row 2, Column 1</td>

      <td>Row 2, Column 2</td>

    </tr>

  </table>

</body>

</html>
```

TABLE CAPTION

The **caption** tag will serve as a title or explanation for the table and it shows up at the top of the table. This tag is deprecated in newer version of HTML/XHTML.

Example

```
<html>

<head>

  <title>HTML Table Caption</title>

</head>

<body>

  <table border = "1" width = "100%">

    <caption>This is the caption</caption>
```

```

<tr>

    <td>row 1, column 1</td><td>row 1, columnn 2</td>

</tr>

<tr>

    <td>row 2, column 1</td><td>row 2, columnn 2</td>

</tr>

</table>

</body>

</html>

```

TABLE HEADER, BODY, AND FOOTER

Tables can be divided into three portions – a header, a body, and a foot. The head and foot are rather similar to headers and footers in a word-processed document that remain the same for every page, while the body is the main content holder of the table.

The three elements for separating the head, body, and foot of a table are –

- **<thead>** – to create a separate table header.
- **<tbody>** – to indicate the main body of the table.
- **<tfoot>** – to create a separate table footer.

Example

```

<html>

<head>

    <title>HTML Table</title>

</head>

<body>

    <table border = "1" width = "100%">

        <thead>

            <tr>

                <td colspan = "4">This is the head of the table</td>

            </tr>

```

```
</thead>

<tfoot>

  <tr>

    <td colspan = "4">This is the foot of the table</td>

  </tr>

</tfoot>

<tbody>

  <tr>

    <td>Cell 1</td>

    <td>Cell 2</td>

    <td>Cell 3</td>

    <td>Cell 4</td>

  </tr>

</tbody>

</table>

</body>

</html>
```

NESTED TABLES

You can use one table inside another table. Not only tables you can use almost all the tags inside table data tag <td>.

Example

Following is the example of using another table and other tags inside a table cell.

```
<html>

  <head>

    <title>HTML Table</title>

  </head>

  <body>
```

```

<table border = "1" width = "100%">

  <tr>

    <td>

      <table border = "1" width = "100%">

        <tr>

          <th>Name</th>

          <th>Salary</th>

        </tr>

        <tr>

          <td>Ramesh Raman</td>

          <td>5000</td>

        </tr>

        <tr>

          <td>Shabbir Hussein</td>

          <td>7000</td>

        </tr>

      </table>

    </td>

  </tr>

</table>

</body>

</html>

```

HTML - LISTS

HTML offers web authors three ways for specifying lists of information. All lists must contain one or more list elements. Lists may contain –

- **** – An unordered list. This will list items using plain bullets.
- **** – An ordered list. This will use different schemes of numbers to list your items.

- **<dl>** – A definition list. This arranges your items in the same way as they are arranged in a dictionary.

HTML UNORDERED LISTS

An unordered list is a collection of related items that have no special order or sequence. This list is created by using HTML **** tag. Each item in the list is marked with a bullet.

Example

```
<html>

<head>

  <title>HTML Unordered List</title>

</head>

<body>

  <ul>

    <li>Beetroot</li>

    <li>Ginger</li>

    <li>Potato</li>

    <li>Radish</li>

  </ul>

</body>

</html>
```

- Beetroot
- Ginger
- Potato
- Radish

THE TYPE ATTRIBUTE

You can use **type** attribute for **** tag to specify the type of bullet you like. By default, it is a disc. Following are the possible options –

```
<ul type = "square">
<ul type = "disc">
<ul type = "circle">
```

Example

Following is an example where we used `<ul type = "square">`

```
<html>

<head>

  <title>HTML Unordered List</title>

</head>

<body>

  <ul type = "square">

    <li>Beetroot</li>

    <li>Ginger</li>

    <li>Potato</li>

    <li>Radish</li>

  </ul>

</body>

</html>
```

- Beetroot
- Ginger
- Potato
- Radish

Example

Following is an example where we used `<ul type = "circle">` -

```
<html>

<head>

  <title>HTML Unordered List</title>

</head>
```

```
<body>

  <ul type = "circle">

    <li>Beetroot</li>

    <li>Ginger</li>

    <li>Potato</li>

    <li>Radish</li>

  </ul>

</body>

</html>
```

- Beetroot
- Ginger
- Potato
- Radish

HTML ORDERED LISTS

If you are required to put your items in a numbered list instead of bulleted, then HTML ordered list will be used. This list is created by using **** tag. The numbering starts at one and is incremented by one for each successive ordered list element tagged with .

Example

```
<html>

<head>

  <title>HTML Ordered List</title>

</head>

<body>

  <ol>

    <li>Beetroot</li>

    <li>Ginger</li>

    <li>Potato</li>
```

```
<li>Radish</li>

</ol>

</body>

</html>
```

1. Beetroot
2. Ginger
3. Potato
4. Radish

THE TYPE ATTRIBUTE

You can use **type** attribute for `<ol>` tag to specify the type of numbering you like. By default, it is a number. Following are the possible options –

```
<ol type = "1"> - Default-Case Numerals.
<ol type = "I"> - Upper-Case Numerals.
<ol type = "i"> - Lower-Case Numerals.
<ol type = "A"> - Upper-Case Letters.
<ol type = "a"> - Lower-Case Letters.
```

Example

Following is an example where we used `<ol type = "1">`

```
<html>

<head>

  <title>HTML Ordered List</title>

</head>

<body>

  <ol type = "1">

    <li>Beetroot</li>

    <li>Ginger</li>

    <li>Potato</li>

    <li>Radish</li>
```

```
</ol>

</body>

</html>
```

1. Beetroot
2. Ginger
3. Potato
4. Radish

Example

Following is an example where we used `<ol type = "I">`

```
<html>

<head>

  <title>HTML Ordered List</title>

</head>

<body>

  <ol type = "I">

    <li>Beetroot</li>

    <li>Ginger</li>

    <li>Potato</li>

    <li>Radish</li>

  </ol>

</body>

</html>
```

- I. Beetroot
- II. Ginger
- III. Potato
- IV. Radish

Example

Following is an example where we used `<ol type = "i">`

```
<html>

<head>

  <title>HTML Ordered List</title>

</head>

<body>

  <ol type = "i">

    <li>Beetroot</li>

    <li>Ginger</li>

    <li>Potato</li>

    <li>Radish</li>

  </ol>

</body>

</html>
```

- i. Beetroot
- ii. Ginger
- iii. Potato
- iv. Radish

Example

Following is an example where we used <ol type = "A" >

```
<html>

<head>

  <title>HTML Ordered List</title>

</head>

<body>

  <ol type = "A">

    <li>Beetroot</li>
```

```
<li>Ginger</li>

<li>Potato</li>

<li>Radish</li>

</ol>

</body>

</html>
```

- A. Beetroot
- B. Ginger
- C. Potato
- D. Radish

Example

Following is an example where we used `<ol type = "a">`

```
<html>

<head>

  <title>HTML Ordered List</title>

</head>

<body>

  <ol type = "a">

    <li>Beetroot</li>

    <li>Ginger</li>

    <li>Potato</li>

    <li>Radish</li>

  </ol>

</body>

</html>
```

- a. Beetroot
- b. Ginger
- c. Potato
- d. Radish

Example

Following is an example where we used `<ol type = "i" start = "4" >`

```
<html>

<head>

  <title>HTML Ordered List</title>

</head>

<body>

  <ol type = "i" start = "4">

    <li>Beetroot</li>

    <li>Ginger</li>

    <li>Potato</li>

    <li>Radish</li>

  </ol>

</body>

</html>
```

- iv. Beetroot
- v. Ginger
- vi. Potato
- vii. Radish

HTML DEFINITION LISTS

HTML and XHTML supports a list style which is called **definition lists** where entries are listed like in a dictionary or encyclopedia. The definition list is the ideal way to present a glossary, list of terms, or other name/value list.

Definition List makes use of following three tags.

- <dl> – Defines the start of the list
- <dt> – A term
- <dd> – Term definition
- </dl> – Defines the end of the list

Example



HTML

This stands for Hyper Text Markup Language

HTTP

This stands for Hyper Text Transfer Protocol

HTML - TEXT LINKS

LINKING DOCUMENTS

A link is specified using HTML tag <a>. This tag is called **anchor tag** and anything between the opening <a> tag and the closing tag becomes part of the link and a user can click that part to reach to the linked document. Following is the simple syntax to use <a> tag.

```
<a href = "Document URL" ... attributes-list>Link Text</a>
```

Example

Let's try following example which links <http://www.tutorialspoint.com> at your page –

```
<html>

<head>

  <title>Hyperlink Example</title>

</head>

<body>

  <p>Click following link</p>

  <a href = "https://www.tutorialspoint.com" target = "_self">Tutorials Point</a>

</body>
```

```
</html>
```

THE TARGET ATTRIBUTE

We have used **target** attribute in our previous example. This attribute is used to specify the location where linked document is opened. Following are the possible options –

Sr.No	Option & Description
1	_blank Opens the linked document in a new window or tab.
2	_self Opens the linked document in the same frame.
3	_parent Opens the linked document in the parent frame.
4	_top Opens the linked document in the full body of the window.
5	targetframe Opens the linked document in a named <i>targetframe</i> .

Example

Try following example to understand basic difference in few options given for target attribute.

```
<html>  
  
<head>
```

```
<title>Hyperlink Example</title>

<base href = "https://www.tutorialspoint.com/">

</head>

<body>

<p>Click any of the following links</p>

<a href = "/html/index.htm" target = "_blank">Opens in New</a> |

<a href = "/html/index.htm" target = "_self">Opens in Self</a> |

<a href = "/html/index.htm" target = "_parent">Opens in Parent</a> |

<a href = "/html/index.htm" target = "_top">Opens in Body</a>

</body>

</html>
```

Click any of the following links

[Opens in New](#) | [Opens in Self](#) | [Opens in Parent](#) | [Opens in Body](#)

DOWNLOAD LINKS

You can create text link to make your PDF, or DOC or ZIP files downloadable. This is very simple; you just need to give complete URL of the downloadable file as follows –

```
<html>

<head>

<title>Hyperlink Example</title>

</head>

<body>

<a href = "https://www.tutorialspoint.com/page.pdf">Download PDF File</a>

</body>

</html>
```

This will produce following link and will be used to download a file.

[Download PDF File](#)

HTML - BACKGROUNDS

By default, your webpage background is white in color. You may not like it, but no worries. HTML provides you following two good ways to decorate your webpage background.

- HTML Background with Colors
- HTML Background with Images

HTML BACKGROUND WITH COLORS

The **bgcolor** attribute is used to control the background of an HTML element, specifically page body and table backgrounds.

Note – The *bgcolor* attribute deprecated in HTML5. Do not use this attribute.

HTML CSS

CSS can be added to HTML elements in 3 ways:

- **Inline**
- **Internal**
- **External**

Inline CSS

An inline CSS is used to apply a unique style to a single HTML element.

An inline CSS uses the style attribute of an HTML element.

Example : `<h1 style="background-color:pink;">Heading </h1>`

Internal CSS

An internal CSS is used to define a style for a single HTML page.

An internal CSS is defined in the `<head>` section of an HTML page, within a `<style>` element:

Example

```
<!DOCTYPE html>
<html>
<head>
<style>
body {background-color: yellow;}
p {color: red;}
</style>
</head>
<body>
<p> An internal CSS is used to define a style for a single HTML page.</p>
</body>
</html>
```

External CSS

An external style sheet is used to define the style for many HTML pages.

With an external style sheet, you can change the look of an entire web site, by changing one file!

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="file1.css">
</head>
<body>
<h1>This is a heading</h1>
<p>This is a paragraph.</p>
</body>
</html>
```

FILE1.CSS

```
body {
  background-color: pink;
}
h1 {
  color: red;
}
p {
```

```
color: yellow;  
}
```

CSS Border

The CSS border property defines a border around an HTML element:

Example

```
p {  
  border: 1px solid cyan;  
}
```

CSS Padding

The CSS padding property defines a padding (space) between the text and the border

Example

```
p {  
  border: 1px solid powderblue;  
  padding
```

CSS Margin

The CSS margin property defines a margin (space) outside the border:

Example

```
p {  
  border: 1px solid powderblue;  
  margin: 50px;  
}
```

ID ATTRIBUTE

To define a specific style for one special element

Use .(hash) before id name when using in style tag.

```
#id_name{  
  
....  
  
}
```

CLASS ATTRIBUTE

Used to apply same format to different elements.

```
.class_name{  
  
....  
  
}
```

External style sheets

External style sheets can be referenced with a full URL or with a path relative to the current web page.

Example

```
<link rel="stylesheet" href="https://www.google.com/html/styles.css">
```

This example links to a style sheet located in the html folder on the current web site:

Example

```
<link rel="stylesheet" href="/html/styles.css">
```

This example links to a style sheet located in the same folder as the current page:

Example

```
<link rel="stylesheet" href="styles.css">
```

HTML - COLORS

Colors are very important to give a good look and feel to your website. You can specify colors on page level using <body> tag or you can set colors for individual tags using **bgcolor** attribute.

The <body> tag has following attributes which can be used to set different colors –

- **bgcolor** – sets a color for the background of the page.
- **text** – sets a color for the body text.
- **alink** – sets a color for active links or selected links.
- **link** – sets a color for linked text.
- **vlink** – sets a color for *visited links* – that is, for linked text that you have already clicked on.

HTML COLOR CODING METHODS

There are following three different methods to set colors in your web page –

- **Color names** – You can specify color names directly like green, blue or red.
- **Hex codes** – A six-digit code representing the amount of red, green, and blue that makes up the color.
- **Color decimal or percentage values** – This value is specified using the `rgb( )` property.

Now we will see these coloring schemes one by one.

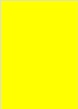
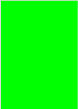
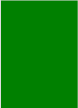
HTML COLORS - COLOR NAMES

You can specify direct a color name to set text or background color. W3C has listed 16 basic color names that will validate with an HTML validator but there are over 200 different color names supported by major browsers.

Note – Check a complete list of [HTML Color Name](#).

W3C STANDARD 16 COLORS

Here is the list of W3C Standard 16 Colors names and it is recommended to use them.

	Black		Gray		Silver		White
	Yellow		Lime		Aqua		Fuchsia
	Red		Green		Blue		Purple
	Maroon		Olive		Navy		Teal

Example

Here are the examples to set background of an HTML tag by color name –

```
<html>

<head>

  <title>HTML Colors by Name</title>

</head>

<body text = "blue" bgcolor = "green">

  <p>Use different color names for for body and table and see the result.</p>

  <table bgcolor = "black">

    <tr>

      <td>

        <font color = "white">This text will appear white on black background.</font>

      </td>

    </tr>

  </table>

</body>

</html>
```

HTML COLORS - HEX CODES

A hexadecimal is a 6 digit representation of a color. The first two digits(RR) represent a red value, the next two are a green value(GG), and the last are the blue value(BB).

A hexadecimal value can be taken from any graphics software like Adobe Photoshop, Paintshop Pro or MS Paint.

Each hexadecimal code will be preceded by a pound or hash sign #. Following is a list of few colors using hexadecimal notation.

Color	Color HEX
	#000000
	#FF0000
	#00FF00
	#0000FF
	#FFFF00
	#00FFFF
	#FF00FF
	#COCOCO
	#FFFFFF

Example

Here are the examples to set background of an HTML tag by color code in hexadecimal –

```

<html>

<head>

  <title>HTML Colors by Hex</title>

</head>

<body text = "#0000FF" bgcolor = "#00FF00">

  <p>Use different color hexa for for body and table and see the result.</p>

  <table bgcolor = "#000000">

    <tr>

      <td>

        <font color = "#FFFFFF">This text will appear white on black background.</font>

      </td>

    </tr>

  </table>

</body>

</html>

```

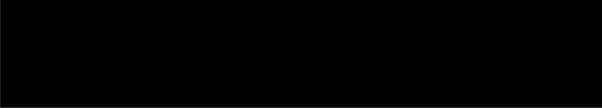
HTML COLORS - RGB VALUES

This color value is specified using the **rgb()** property. This property takes three values, one each for red, green, and blue. The value can be an integer between 0 and 255 or a percentage.

Note – All the browsers does not support rgb() property of color so it is recommended not to use it.

Following is a list to show few colors using RGB values.

Color	Color RGB

	rgb(0,0,0)
	rgb(255,0,0)
	rgb(0,255,0)
	rgb(0,0,255)
	rgb(255,255,0)
	rgb(0,255,255)
	rgb(255,0,255)
	rgb(192,192,192)
	rgb(255,255,255)

Example

Here are the examples to set background of an HTML tag by color code using rgb() values

```
<html>

<head>

  <title>HTML Colors by RGB code</title>

</head>

<body text = "rgb(0,0,255)" bgcolor = "rgb(0,255,0)">

  <p>Use different color code for for body and table and see the result.</p>

  <table bgcolor = "rgb(0,0,0)">

    <tr>

      <td>
```

```

<font color = "rgb(255,255,255)">This text will appear white on black background.</font>

</td>

</tr>

</table>

</body>

</html>

```

BROWSER SAFE COLORS

Here is the list of 216 colors which are supposed to be safest and computer independent colors. These colors vary from hexa code 000000 to FFFFFFFF and they will be supported by all the computers having 256 color palette.

000000	000033	000066	000099	0000CC	0000FF
003300	003333	003366	003399	0033CC	0033FF
006600	006633	006666	006699	0066CC	0066FF
009900	009933	009966	009999	0099CC	0099FF
00CC00	00CC33	00CC66	00CC99	00CCCC	00CCFF
00FF00	00FF33	00FF66	00FF99	00FFCC	00FFFF
330000	330033	330066	330099	3300CC	3300FF
333300	333333	333366	333399	3333CC	3333FF
336600	336633	336666	336699	3366CC	3366FF
339900	339933	339966	339999	3399CC	3399FF

33CC00	33CC33	33CC66	33CC99	33CCCC	33CCFF
33FF00	33FF33	33FF66	33FF99	33FFCC	33FFFF
660000	660033	660066	660099	6600CC	6600FF
663300	663333	663366	663399	6633CC	6633FF
666600	666633	666666	666699	6666CC	6666FF
669900	669933	669966	669999	6699CC	6699FF
66CC00	66CC33	66CC66	66CC99	66CCCC	66CCFF
66FF00	66FF33	66FF66	66FF99	66FFCC	66FFFF
990000	990033	990066	990099	9900CC	9900FF
993300	993333	993366	993399	9933CC	9933FF
996600	996633	996666	996699	9966CC	9966FF
999900	999933	999966	999999	9999CC	9999FF
99CC00	99CC33	99CC66	99CC99	99CCCC	99CCFF
99FF00	99FF33	99FF66	99FF99	99FFCC	99FFFF
CC0000	CC0033	CC0066	CC0099	CC00CC	CC00FF

CC3300	CC3333	CC3366	CC3399	CC33CC	CC33FF
CC6600	CC6633	CC6666	CC6699	CC66CC	CC66FF
CC9900	CC9933	CC9966	CC9999	CC99CC	CC99FF
CCCC00	CCCC33	CCCC66	CCCC99	CCCCCC	CCCCFF
CCFF00	CCFF33	CCFF66	CCFF99	CCFFCC	CCFFFF
FF0000	FF0033	FF0066	FF0099	FF00CC	FF00FF
FF3300	FF3333	FF3366	FF3399	FF33CC	FF33FF
FF6600	FF6633	FF6666	FF6699	FF66CC	FF66FF
FF9900	FF9933	FF9966	FF9999	FF99CC	FF99FF
FFCC00	FFCC33	FFCC66	FFCC99	FFCCCC	FFCCFF
FFFF00	FFFF33	FFFF66	FFFF99	FFFFCC	FFFFFF

HTML - MARQUEES

SYNTAX

A simple syntax to use HTML <marquee> tag is as follows –

```
<marquee attribute_name = "attribute_value"....more attributes>
  One or more lines or text message or image
</marquee>
```

THE <MARQUEE> TAG ATTRIBUTES

Following is the list of important attributes which can be used with <marquee> tag.

Sr.No	Attribute & Description
1	width This specifies the width of the marquee. This can be a value like 10 or 20% etc.
2	height This specifies the height of the marquee. This can be a value like 10 or 20% etc.
3	direction This specifies the direction in which marquee should scroll. This can be a value like <i>up</i> , <i>down</i> , <i>left</i> or <i>right</i> .
4	behavior This specifies the type of scrolling of the marquee. This can have a value like <i>scroll</i> , <i>slide</i> and <i>alternate</i> .
5	scrolldelay This specifies how long to delay between each jump. This will have a value like 10 etc.
6	scrollamount

	This specifies the speed of marquee text. This can have a value like 10 etc.
--	--

Examples - 1

```
<html>

<head>

  <title>HTML marquee Tag</title>

</head>

<body>

  <marquee>This is basic example of marquee</marquee>

</body>

</html>
```

Examples - 2

```
<html>

<head>

  <title>HTML marquee Tag</title>

</head>

<body>

  <marquee width = "50%">This example will take only 50% width</marquee>

</body>

</html>
```

Examples - 3

```
<html>

<head>

  <title>HTML marquee Tag</title>

</head>
```

```
<body>

  <marquee direction = "right">This text will scroll from left to right</marquee>

</body>
</html>
```

Examples - 4

```
<html>

<head>

  <title>HTML marquee Tag</title>

</head>

<body>

  <marquee direction = "up">This text will scroll from bottom to up</marquee>

</body>
</html>
```

HTML Forms

Form tag

Example: The HTML <form> element defines a form that is used to collect user input:

Syntax : <form> *form elements* </form>

- <input>
- <textarea>
- <button>
- <select>
- <option>
- <optgroup>
- <fieldset>
- <label>

Form elements are different types of input elements, like text fields, checkboxes, radio buttons, submit buttons, and more.

Form attributes:

Input Tag

- `<input>` specifies an input field where user can enter data.
- There is no closing tag for input tag.

Attributes name	Values	description
Type	Text Password Email Radio checkbox button submit File Hidden Range url color date datetime-local search etc..	Specifies the type <code><input></code> element to display.
Value	<i>Your text</i>	Specifies the value of the input field.
Placeholder	<i>Your text</i>	Specifies the value if the input element is not focused.
Name	<i>Name</i>	Specifies the name of an <code><input></code> element.
Readonly	Readonly/ <i>no value</i>	Specifies that the input field is read only.

Required	Required/ <i>no value</i>	Specifies that the input field must be filled out.
Autofocus	Autofocus/ <i>no value</i>	Specifies that the input field must be focused automatically.
Checked	Checked/ <i>no value</i>	Specifies that the input element should be pre- selected(for 'radio' and 'checkbox')

Example:

1. `<input type="text" placeholder="Name" required> <br>`
2. `<input type="email" value="@ " autofocus> <br>`
3. `<input type="password" placeholder="Password"> <br>`
4. `<input type="radio" name="gender">Male <input type="radio" name="gender"> female <br>`
5. `<input type="checkbox" name="ha">Hacker <br>`
6. `<input type="checkbox" name="pro">Programmer<br>`
7. `<input type="file"><br>`
8. `<input type="submit" value="submit">`

Select Tag

- `<select>` element is used to create a drop down list.

Attributes name	Values	Description
Multiple	Multiple/ <i>no value</i>	Specifies that multiple option can be selected.
Required	Required/ <i>no value</i>	Must be filled out.
Disable	Disable/ <i>no value</i>	Specifies that the drop down is disabled.
Autofocus	Autofocus/ <i>no value</i>	Specifies that the drop down list should be focused automatically.
Name	<i>name</i>	Defines a name for the drop-down list.
size	<i>numbers</i>	the number of visible options in a drop-down list.

Example

```

<form >
<select name="cars" multiple>
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="opel">Opel</option>
  <option value="audi">Audi</option>
</select>
<select name="cars" disabled>
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="opel">Opel</option>
  <option value="audi">Audi</option>
</select>
<select name="cars" autofocus>
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="opel">Opel</option>
  <option value="audi">Audi</option>
</select>
<select name="cars" required>
  <option value="">none</option>
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="opel">Opel</option>
  <option value="audi">Audi</option>
</select>

```

```

<select name="cars" size="2">
  <option value="">none</option>
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="opel">Opel</option>
  <option value="audi">Audi</option>
</select>
<input type="submit">
</form>

```

Option Tag

- There is <option> tag that is used with the <select> tag.

Attributes name	Values	Description
Disable	Disable/ <i>no value</i>	Specifies that the option is disabled.
Selected	Selected/ <i>no value</i>	Specifies that the option is pre-Selected when page loads. (chosen).
Value	<i>Your value</i>	Specifies the value.

Example:

```

<select>

  <option value="hacker">Hacker</option>

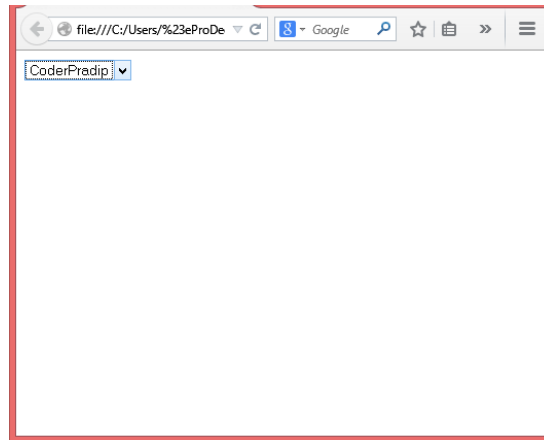
  <option value="coderpradip" selected>CoderPradip</option>

  <option value="doctor" disabled>Doctor</option>

  <option value="programmer">Programmer</option>

</select>

```



Optgroup Tag

- The <optgroup> is used to group related options in a drop-down list.
-

Attribute	Value	Description
disabled	disabled	Specifies that an option-group should be disabled
label	text	Specifies a label for an option-group

Example

```
<!DOCTYPE html>
<html>
<body>
<form >
  <select name="foods" required>
    <option value="">Choose any fod</option>
    <optgroup label="Ice cream">
      <option value="venilla"> Venilla </option>
      <option value="chocolate"> Chocolate </option>
    </optgroup>
```

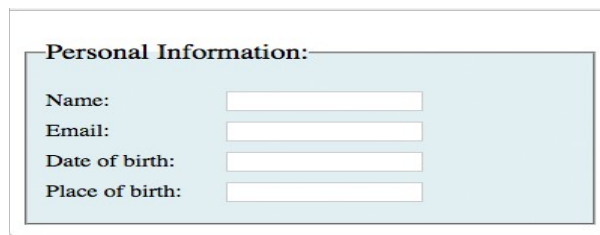
```

<optgroup label="Dairymilk">
  <option value=" Silk ">Silk</option>
  <option value=" Fruitnut ">Fruitnut</option>
</optgroup>
</select>
<input type="submit">
</form>
</body>
</html>

```

Fieldset Tag

- The <fieldset> tag is used to group related elements in a form. It draws a box around the related elements.



Example

```

<!DOCTYPE html>
<html>
<body>
<form>
<fieldset>
  <legend> Bio-data</legend>
  Name: <input type="text"><br>
  Email: <input type="email"><br>
  Date of birth: <input type="text">
</fieldset>
</form>
</body>
</html>

```

Attribute	Value	Description
disabled	disabled	Specifies that a group of related form elements should be disabled

form	<i>form_id</i>	Specifies one or more forms the fieldset belongs to
name	<i>text</i>	Specifies a name for the fieldset

<label> tag

The <label> element does not render as anything special for the user. However, it provides a usability improvement for mouse users, because if the user clicks on the text within the <label> element, it toggles the control.

The for attribute of the <label> tag should be equal to the id attribute of the related element to bind them together.

EXAMPLE

```
<form action="/action_page.php">
  <label for="male">Male</label>
  <input type="radio" name="gender" id="male" value="male"><br>
  <label for="female">Female</label>
  <input type="radio" name="gender" id="female" value="female"><br>
  <label for="other">Other</label>
  <input type="radio" name="gender" id="other" value="other"><br><br>
  <input type="submit" value="Submit">
</form>
```

Attribute	Value	Description
for	<i>element_id</i>	Specifies which form element a label is bound to
form	<i>form_id</i>	Specifies one or more forms the label belongs to

```
<form action="FILE.htm" method="post" id="form1">
```

```

<input type="text" name="name" id="male">
<label for="female">password</label>
<input type="password" name="name" id="female">
<button value="submit">Submit</button>
</form>
<label for="male" form="form1">Firstname</label>

```

FORM TAG

form action attribute

The action attribute specifies where to send the form-data when a form is submitted.

Example

```

<form action="URL">
<form action="/action_page.php" method="get">
  First name: <input type="text" name="fname"><br>
  Last name: <input type="text" name="lname"><br>
  <input type="submit" value="Submit">
</form>

```

form method attribute

method="get, post"

The form-data can be sent as URL variables (with method="get") or as HTTP post transaction (with method="post").

Example

```

<form action="URL">
<form action="/action_page.php" method="get">
  First name: <input type="text" name="fname"><br>
  Last name: <input type="text" name="lname"><br>
  <input type="submit" value="Submit">
</form>

```

form novalidate attribute

(**validation**-- First of all, the form must be checked to make sure all the mandatory fields are filled in. It would require just a loop through each field in the form and check for data.)

If validation attribute is used, the Form doesn't validate.

Example

```
<form action="file:///C:/Users/Kice/Desktop/HTML/nothing.htm" method="post"
novalidate >
<label for="male">Firstname</label>
<input type="text" name="name" id="male" required>
<label for="female">password</label>
<input type="password" name="name" id="female">
<button value="submit">Submit</button>
</form>
```

form target attribute

The target attribute specifies a name or a keyword that indicates where to display the response that is received after submitting the form

Value	Description
_blank	The response is displayed in a new window or tab
_self	The response is displayed in the same frame (this is default)
_parent	The response is displayed in the parent frame
_top	The response is displayed in the full body of the window
<i>framename</i>	The response is displayed in a named iframe

Iframe Tag

<iframe> specifies an inline frame.

Attributes name	Values	Description
Src	<i>url/path</i>	Specifies the Path embedded document.
Name	<i>Your text</i>	Specifies the name of Frame.

Height/width	Value	Specifies the height/width of Frame.
--------------	-------	--------------------------------------

Example:

```
<iframe src="mygame.html" name="myframe" height="300" width="300">
```

